

	$\mathcal{K}_{0^{+}}^{\#1}$	$\mathcal{K}_{0^{-}}^{\#1\alpha\beta\chi}$					
$\mathcal{K}_{0^{+}}^{\#1} \dagger$	$-\frac{3\Upsilon_1 k^2}{2}$	0					
$\mathcal{K}_{0^{-}}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	$\mathcal{K}_{1^{+}}^{\#1\alpha\beta}$	$\mathcal{K}_{1^{+}}^{\#2\alpha\beta}$	$\mathcal{K}_{1^{-}}^{\#1\alpha}$	$\mathcal{K}_{1^{-}}^{\#2\alpha}$	
	$\mathcal{K}_{1^{+}}^{\#1} \dagger^{\alpha\beta}$	$-k^2 \frac{{}^{(4)}\kappa_3}{\sqrt{2}}$	$\frac{k^2 \frac{{}^{(4)}\kappa_4}{2}}{2\sqrt{2}}$	0	0		
	$\mathcal{K}_{1^{+}}^{\#2} \dagger^{\alpha\beta}$	$\frac{k^2 \frac{{}^{(4)}\kappa_4}{2}}{2\sqrt{2}}$	$-\frac{\Upsilon_2 k^2}{2}$	0	0		
	$\mathcal{K}_{1^{+}}^{\#1} \dagger^{\alpha}$	0	0	$-k^2 \frac{{}^{(4)}\kappa_2}{\sqrt{2}}$	$-\sqrt{2} k^2 \frac{{}^{(4)}\kappa_2}{\sqrt{2}}$		
	$\mathcal{K}_{1^{+}}^{\#2} \dagger^{\alpha}$	0	0	$-\sqrt{2} k^2 \frac{{}^{(4)}\kappa_2}{\sqrt{2}}$	$-2 k^2 \frac{{}^{(4)}\kappa_2}{\sqrt{2}}$	$\mathcal{K}_{2^{+}}^{\#1\alpha\beta}$	$\mathcal{K}_{2^{-}}^{\#1\alpha\beta\chi}$
				$\mathcal{K}_{2^{+}}^{\#1} \dagger^{\alpha\beta}$	0	0	
				$\mathcal{K}_{2^{-}}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	

	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^+}^{\#1\alpha\beta\chi}$						
$\mathcal{J}_{0^+}^{\#1} \dagger$	$\frac{1}{\text{Det}(0^+)}$	0						
$\mathcal{J}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	$\mathcal{J}_{1^+}^{\#1\alpha\beta}$	$\mathcal{J}_{1^+}^{\#2\alpha\beta}$	$\mathcal{J}_{1^+}^{\#1\alpha}$	$\mathcal{J}_{1^+}^{\#2\alpha}$		
$\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$-\frac{\Upsilon_2 k^2}{2 \text{Det}(1^+)}$	$-\frac{k^2 \overset{(4)}{\kappa}_4}{2 \sqrt{2} \text{Det}(1^+)}$	0	0				
$\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$-\frac{k^2 \overset{(4)}{\kappa}_4}{2 \sqrt{2} \text{Det}(1^+)}$	$-\frac{k^2 \overset{(4)}{\kappa}_3}{\text{Det}(1^+)}$	0	0				
$\mathcal{J}_{1^+}^{\#1} \dagger^\alpha$	0	0	$-\frac{1}{9 k^2 \overset{(4)}{\kappa}_2}$	$-\frac{\sqrt{2}}{9 k^2 \overset{(4)}{\kappa}_2}$				
$\mathcal{J}_{1^+}^{\#2} \dagger^\alpha$	0	0	$-\frac{\sqrt{2}}{9 k^2 \overset{(4)}{\kappa}_2}$	$-\frac{2}{9 k^2 \overset{(4)}{\kappa}_2}$	$\mathcal{J}_{2^+}^{\#1\alpha\beta}$	$\mathcal{J}_{2^+}^{\#1\alpha\beta\chi}$		
			$\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta}$	0	0			
			$\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	0			

Abbreviations used in matrices			
$\Upsilon_1 == \frac{{}^{(4)}\kappa_1}{{}^{(4)}\kappa_2} + \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_3} \&\& \Upsilon_2 == 9 \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_3} - \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_3} \&\&$ $\text{Det}(0^+) == -\frac{3}{2} k^2 (\frac{{}^{(4)}\kappa_1}{{}^{(4)}\kappa_2} + \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_3}) \&\& \text{Det}(1^+) == \frac{1}{8} k^4 (36 \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_3} \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_3} - (2 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} + \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4})^2)$			
Lagrangian			
$\frac{{}^{(4)}\kappa_1}{6} \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^{\chi\alpha\beta} \mathcal{K}^\alpha_{\alpha\beta} + \frac{{}^{(4)}\kappa_2}{2} \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^{\chi\alpha\beta} \mathcal{K}^\alpha_{\alpha\beta} + \frac{{}^{(4)}\kappa_3}{2} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\alpha\chi} + \frac{{}^{(4)}\kappa_4}{2} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} - \frac{{}^{(4)}\kappa_3}{2} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\chi\alpha} -$ $6 \frac{{}^{(4)}\kappa_2}{\kappa_2} \partial^{\chi\alpha\beta} \mathcal{K}^\delta_{\alpha\beta} \partial_\delta \mathcal{K}^\delta_{\chi\beta} - \frac{9}{2} \frac{{}^{(4)}\kappa_4}{\kappa_2} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \frac{1}{2} \frac{{}^{(4)}\kappa_3}{\kappa_3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \frac{1}{2} \frac{{}^{(4)}\kappa_4}{\kappa_4} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi}$			
Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_0^{\#1\alpha\beta\chi} == 0$	1	$\partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta}$	
$\mathcal{J}_1^{\#1\alpha} == \mathcal{J}_1^{\#2\alpha}$	3	$\partial_\chi \partial^\alpha \mathcal{J}^{\beta\chi\alpha} + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha\chi} == 0$	
$\mathcal{J}_2^{\#1\alpha\beta\chi} == 0$	5	$3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta}_{\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} +$ $4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^{\delta}_{\delta}{}^\epsilon + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\alpha}_{\delta} ==$ $3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha}_{\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} +$ $2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^{\delta}_{\delta}{}^\epsilon + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\beta}_{\delta}$	
$\mathcal{J}_2^{\#1\alpha\beta} == 0$	5	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial^\epsilon \partial_\delta \mathcal{J}^{\chi\delta}_{\chi} == 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^{\chi\delta}_{\chi} + 3 (\partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi})$	
Total # constraint(s):	14		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$-\frac{216 \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} - 24 \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} (2 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} + \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4}) + (2 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} + \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4})^2}{\frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} (36 \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} - (2 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} + \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4})^2)}$
	1	0	$-\frac{48 \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} - 8 \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} (2 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} + \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4}) + (2 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} + \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4})^2}{\frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} (36 \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} - (2 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} + \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4})^2)}$
Resolved unitarity condition(s):		$(\frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4} < 0 \&\& ((\frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} < 0 \&\& (\frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} < \frac{4 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} + 4 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4} + \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4}}{36 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3}} \parallel \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} > 0))) \parallel (\frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} == 0 \&\& \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} > 0) \parallel$ $(0 < \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} < -\frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_2} \&\& 0 < \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} < \frac{4 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} + 4 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4} + \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4}}{36 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3}}) \parallel (\frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} > -\frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_2} \&\& 0 < \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} < \frac{4 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} + 4 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4} + \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4}}{36 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3}}))) \parallel$ $(\frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4} == 0 \&\& ((\frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} < 0 \&\& (\frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} < \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} \parallel \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} > 0))) \parallel (\frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} > 0 \&\& 0 < \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} < \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3}))) \parallel$ $(\frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4} > 0 \&\& ((\frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} < -\frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_2} \&\& (\frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} < \frac{4 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} + 4 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4} + \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4}}{36 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3}} \parallel \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} > 0))) \parallel$ $(\frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} == -\frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_2} \&\& (\frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} < 0 \parallel \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} > 0))) \parallel (-\frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_2} < \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} < 0 \&\& (\frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} < \frac{4 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} + 4 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4} + \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4}}{36 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3}} \parallel \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} > 0))) \parallel$ $(\frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} == 0 \&\& \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} > 0) \parallel ((\frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} > 0 \&\& 0 < \frac{{}^{(4)}\kappa_2}{{}^{(4)}\kappa_2} < \frac{4 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} + 4 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3} \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4} + \frac{{}^{(4)}\kappa_4}{{}^{(4)}\kappa_4}}{36 \frac{{}^{(4)}\kappa_3}{{}^{(4)}\kappa_3}})))$	